IRCTC Tatkal Booking

Business Analysis Project

Tushar Jain
October 2026

Self-made portfolio project. The solution and the business scenario are my own and are not IRCTC's real plans. The problems come from 5,100 real reviews of the IRCTC Rail Connect app on the Google Play Store (5-20 Oct 2026).

Contents

1. Understanding the need
2. Business Requirements Document (BRD)
3. Functional Requirements Document (FRD)
4. Process flow (To-Be)
5. Wireframes
6. Use case diagram
7. User stories
8. UAT test cases

1. Understanding the need

Tatkal tickets are last-minute train tickets. Booking opens one day before the journey, at 10 AM for AC classes and 11 AM for non-AC, and good trains sell out in a few minutes. Most people book them on the IRCTC Rail Connect app, so those first few minutes decide whether you travel or not.

I picked this because almost everyone in India has a Tatkal story. To check it wasn't just my opinion, I pulled the latest 5,100 Play Store reviews of the app (5-20 Oct 2026) and went through the 733 detailed complaints (1-3 stars).

What the reviews show

What people complained about	Reviews	% of complaints
App slow, hangs or 'high load' errors	222	30%
Payment taken but ticket not booked / money stuck	206	28%
Tatkal booking fails when the window opens	157	21%
Ended up on waiting list / no seats	109	15%
Login, OTP or Aadhaar problems	46	6%
Agents or bots getting the tickets	39	5%

A review can be in more than one row. I tagged them by keywords, so the numbers are close, not exact. Overall 25% of all 5,100 reviews were 1 or 2 stars.

Some of the reviews (copied as written)

- "High load error while booking tatkal ticket, app crashing" (1 star, 16 Oct 2026)
- "Issues: After Payment Deduct but PNR Not Generate Just show Error. Resolve it." (2 stars, 20 Oct 2026)
- "tatkal quota kholne ke bad yah app itna slow ho jata hai ki ticket hi nahin nikal pata" (1 star, 20 Oct 2026)

What I noticed

- The biggest pain is the first 2-3 minutes: people get 'high load, please retry', retry, and by the time it works the seats are gone.
- The second is money. People pay, get an 'internal error', and don't know if they have a ticket or when they'll get the money back. The money does come back (within 3 working days), but nobody tells them that on the screen.
- Some things are already fixed by IRCTC rules: since July 2025 only Aadhaar-verified users can book Tatkal online, and agents can't book in the first 30 minutes. So the problem now is mostly the app experience, not bots.

Problem statement

The problem of	the IRCTC app failing at the moment Tatkal opens (high-load errors, crashes, logouts) and not telling people clearly what happened when the payment goes through but the ticket doesn't
affects	passengers, IRCTC customer care, the app product team, and the reservation and payment tech teams
the impact of which is	missed tickets, money stuck for days, angry reviews, a flood of complaints to customer care, and people turning to agents
a successful solution would be	a calm, fair way into the booking at 10 / 11 AM, details saved in advance so booking takes seconds, and a clear booking status with automatic refunds when something fails

2. Business Requirements Document (BRD)

Project	IRCTC Tatkal Booking Experience
Version	1.0
Prepared by	Tushar Jain
Sponsor	Product Head, IRCTC Rail Connect app
Date	21/10/2026
Status	Draft for review

Revision history

Version	Date	Author	Changes
0.1	21/10/2026	Tushar Jain	First draft
1.0	21/10/2026	Tushar Jain	Added review data, KPIs and business rules

Executive summary

Tatkal booking on the app breaks exactly when people need it. In recent reviews, 30% of complaints were about the app being slow or showing 'high load', 28% about money being taken without a ticket and 21% about Tatkal failing outright. This project changes the booking journey in three ways: passengers save everything in advance ('Tatkal Ready'), they wait in a proper queue instead of getting errors, and after paying they always see a clear status, with an automatic refund if the ticket isn't booked. The aim is to halve these complaints within six months.

Background

Lakhs of people try to book Tatkal at the same moment every day. Right now the app either lets you in or throws an error, and people keep retrying, which adds more load. When payment succeeds but the PNR isn't generated, the app shows a generic error. The refund is automatic and comes within 3 working days, but the passenger isn't told that, so they panic, call customer care or post a 1-star review.

Problem statement

The problem of	app failures when Tatkal opens and unclear status when payment succeeds but the ticket doesn't
affects	passengers, customer care, the app product team, reservation and payment tech teams
the impact of which is	missed tickets, money stuck, complaints and bad reviews
a successful solution would be	a fair queue, saved booking details and a clear status with automatic refunds

Business objectives

ID	Objective	Target	When
BO-01	Fewer 'app slow / high load' complaints	From 30% to under 15% of complaints	6 months after launch
BO-02	Fewer 'money taken, no ticket' complaints	From 28% to under 10% of complaints	6 months after launch
BO-03	Faster booking once it's your turn	Tatkal booking done in under 60 seconds after leaving the queue	3 months after launch
BO-04	Every failed booking refunded without a complaint	100% auto-refund started within 1 hour	From launch

Scope

In scope	Out of scope
Tatkal Ready: save journey, passengers and payment in advance	Number of Tatkal seats and quota rules
Waiting room / queue when Tatkal opens	Tatkal and Premium Tatkal fares
Holding the session so people aren't logged out	Aadhaar verification policy itself
Seat re-check before payment	Server and infrastructure upgrade (handled by the tech team)
Clear booking status after payment	General (non-Tatkal) booking
Auto-retry of PNR and automatic refund messages	Agent licensing and agent apps
Customer care screen for failed transactions	IRCTC website (app only in this phase)
Respecting 'book only if confirmed berth'	

Stakeholders

Stakeholder	What they care about	Role
Passengers	Getting a ticket, not losing money	End users
Product Head, Rail Connect app	Fewer complaints, better ratings	Sponsor, signs off
App product team	Building the right changes	Owns the backlog
Reservation (PRS) tech team	Load on the booking system	Builds queue and PNR changes
Payment team / bank partners	Smooth payments and refunds	Refund process
Customer care	Fewer calls about failed bookings	Uses failed transactions screen
Railway Board / policy team	Tatkal rules are followed	Approves rule changes

As-Is process

Step	What happens today	Problem
1	Passenger logs in just before 10 / 11 AM	Gets logged out or OTP is slow
2	Searches the train and fills passenger details at 10:00	Takes time while seats are going
3	App shows 'high load, please retry'	Passenger keeps retrying, more load
4	Passenger pays	Sometimes money is taken but 'internal error' shows
5	No PNR, no clear message	Passenger doesn't know if money will come back
6	Passenger calls customer care or posts a review	Care has no quick view of the failed payment

To-Be process

Step	What will happen	What changed
1	Passenger sets up Tatkal Ready the day before (train, class, passengers, payment)	Nothing to type at 10:00
2	App checks Aadhaar is verified in advance	No surprise at booking time
3	Waiting room opens at 9:58 / 10:58, passenger gets a queue number	No more 'high load' errors
4	When it's their turn, they have 5 minutes to check and pay	Session held, no logouts
5	Seats re-checked before paying	No paying for a WL ticket by mistake
6	After payment the app shows a status: booked, retrying or refund started	Passenger always knows what's going on
7	If no PNR, system retries once, then starts the refund automatically	No complaint needed

The swimlane for this is in section 4.

Business requirements

ID	Requirement	Priority	Source
BR-01	Passengers must be able to save their Tatkal journey, passengers and payment method before the window opens	Must	Reviews (Tatkal fails 21%)
BR-02	Passengers must wait in a fair queue when Tatkal opens instead of getting errors	Must	Reviews (high load 30%)
BR-03	Passengers must not be logged out while waiting or booking	Must	Reviews
BR-04	Passengers must see if seats are still available before paying	Should	Reviews (ended up on WL 15%)
BR-05	After paying, passengers must see a clear booking status	Must	Reviews (money taken 28%)
BR-06	Failed bookings must be refunded automatically and the passenger told when	Must	Reviews, customer care
BR-07	'Book only if confirmed berth' must always be respected	Must	Reviews
BR-08	Customer care must be able to see failed transactions and refund status	Should	Customer care
BR-09	Error messages must say what happened and what to do next	Should	Reviews
BR-10	The operations team could see live Tatkal load	Could	PRS tech team
BR-11	Changing the Tatkal quota or fares	Won't	Out of scope (policy)

Assumptions, constraints and dependencies

Type	ID	Description	Owner
Assumption	A-01	Passengers already have a master passenger list in the app	App team
Assumption	A-02	Most passengers use UPI for Tatkal	Payment team
Constraint	C-01	Must follow current Tatkal rules (Aadhaar-verified users only, agents blocked for the first 30 minutes)	Policy team
Constraint	C-02	Refund for failed transactions within 3 working days (current IRCTC rule)	Payment team
Constraint	C-03	Max 4 passengers per Tatkal PNR	Policy team
Dependency	D-01	PRS team to support a queue in front of booking	PRS tech team
Dependency	D-02	Payment gateway to send success / failure in real time	Payment team

Risks

ID	Risk	Impact	Mitigation
R-01	People think the queue is unfair	High	Show queue number and wait time; first-come order
R-02	People use many phones to jump the queue	Medium	One active Tatkal session per account
R-03	Queue itself goes down at 10:00	High	Load test before launch, fall back to current flow
R-04	Refunds still take days because of banks	Medium	Show expected date clearly; care can retry failed refunds

KPIs

KPI	Now	Target	BO
Share of complaints about slow / high load	30%	Under 15%	BO-01
Share of complaints about money taken, no ticket	28%	Under 10%	BO-02
Time to book after leaving the queue	Not measured	Under 60 sec	BO-03
Failed bookings with auto-refund started within 1 hour	Not measured	100%	BO-04

'Now' numbers are from my Play Store sample (Oct 2026). In a real project I'd replace them with IRCTC's own data.

Glossary

Term	Meaning
Tatkal	Last-minute quota that opens one day before the journey
PNR	Ticket reference number
WL	Waiting list
PRS	Passenger Reservation System (the booking back end)
Tatkal Ready	New feature to save booking details in advance
Waiting room	Queue screen shown when Tatkal opens

Sign-off

Name	Role	Signature	Date
	Product Head, Rail Connect (sponsor)		
	PRS tech team lead		
	Payment team lead		
	Customer care head		

3. Functional Requirements Document (FRD)

Project	IRCTC Tatkal Booking Experience
Version	1.0
Prepared by	Tushar Jain
Based on	BRD v1.0
Date	21/10/2026

Purpose

This explains how the new Tatkal booking flow will work in the app, so the product team, developers and testers know exactly what to build. Scope is the same as the BRD.

User roles

Role	**Can do**
Passenger	Set up Tatkal Ready, join the queue, book, pay, see booking and refund status
Customer care agent	See failed transactions, check refund status, retry a failed refund, resend refund SMS
Operations team	Watch live Tatkal load and queue length
Admin	Change queue settings (opening time, session time)

Functional requirements

ID	The system shall...	BR	Priority
FR-01	let the passenger save train, date, class, quota and payment method as 'Tatkal Ready'	BR-01	Must
FR-02	let the passenger pick up to 4 passengers from the master list	BR-01	Must
FR-03	check Aadhaar verification when Tatkal Ready is saved and ask to verify if not done	BR-01	Must
FR-04	open a waiting room 2 minutes before Tatkal opens and show queue position and estimated wait	BR-02	Must
FR-05	let passengers into booking in the order they joined the queue	BR-02	Must
FR-06	keep the session active while in the queue and give 5 minutes to pay once it's their turn	BR-03	Must
FR-07	re-check seat availability before payment and warn if it has changed	BR-04	Should
FR-08	show 'Payment failed, nothing deducted' and allow retry when payment fails	BR-05, BR-09	Must
FR-09	show a booking status screen after payment (payment received, confirming, booked)	BR-05	Must
FR-10	retry PNR generation once and, if it still fails, start the refund and notify the passenger	BR-06	Must
FR-11	not issue a ticket when 'book only if confirmed berth' is on and no berth is confirmed, and refund in full	BR-07	Must
FR-12	show customer care a list of failed transactions with refund status	BR-08	Should
FR-13	let customer care retry a failed refund and log every action	BR-08	Should
FR-14	replace generic errors with clear messages that say what to do	BR-09	Should
FR-15	show the operations team live queue length and booking rate	BR-10	Could

Business rules

RULE-01: Max 4 passengers per Tatkal PNR.

RULE-02: Only Aadhaar-verified accounts can book Tatkal.

RULE-03: Waiting room opens at 9:58 AM (AC) and 10:58 AM (non-AC).

RULE-04: Queue is first come, first served from the moment the passenger joins.

RULE-05: Session is held in the queue; once it's their turn the passenger has 5 minutes to pay.

RULE-06: If 'book only if confirmed berth' is ticked and no confirmed berth is available, no ticket is booked and the full amount is refunded.

RULE-07: If no PNR within 2 minutes of payment, retry once.

RULE-08: Refund for a failed booking starts within 1 hour and is credited within 3 working days.

RULE-09: Only one active Tatkal session per account.

RULE-10: Authorised agents can't book in the first 30 minutes (existing rule).

Screens

ID	Screen	Linked to
WF-01	Tatkal Ready	FR-01, FR-02, FR-03
WF-02	Waiting room	FR-04, FR-05, FR-06
WF-03	Review and pay	FR-06, FR-07, FR-08
WF-04	Booking status	FR-09, FR-10, FR-11
WF-05	Failed transactions (customer care)	FR-12, FR-13

Data requirements

Data	Field	Rule
Tatkal Ready	Train no., date, class, quota	Date must be tomorrow; quota Tatkal or Premium Tatkal
Tatkal Ready	Passengers	1 to 4, from master list
Tatkal Ready	Payment method	Saved UPI ID or card
Queue entry	Queue number, join time	System generated
Transaction	Txn ID, amount, payment status	From payment gateway
Transaction	PNR status	Booked / Retrying / Failed
Refund	Refund ref, start time, expected date, status	Started / Credited / Failed

Notifications

When	To	Channel
Tatkal Ready saved (reminder 10 min before opening)	Passenger	Push
Ticket booked	Passenger	Push, SMS, email
Refund started (with expected date)	Passenger	Push, SMS
Refund failed	Customer care	Care console

Non-functional requirements

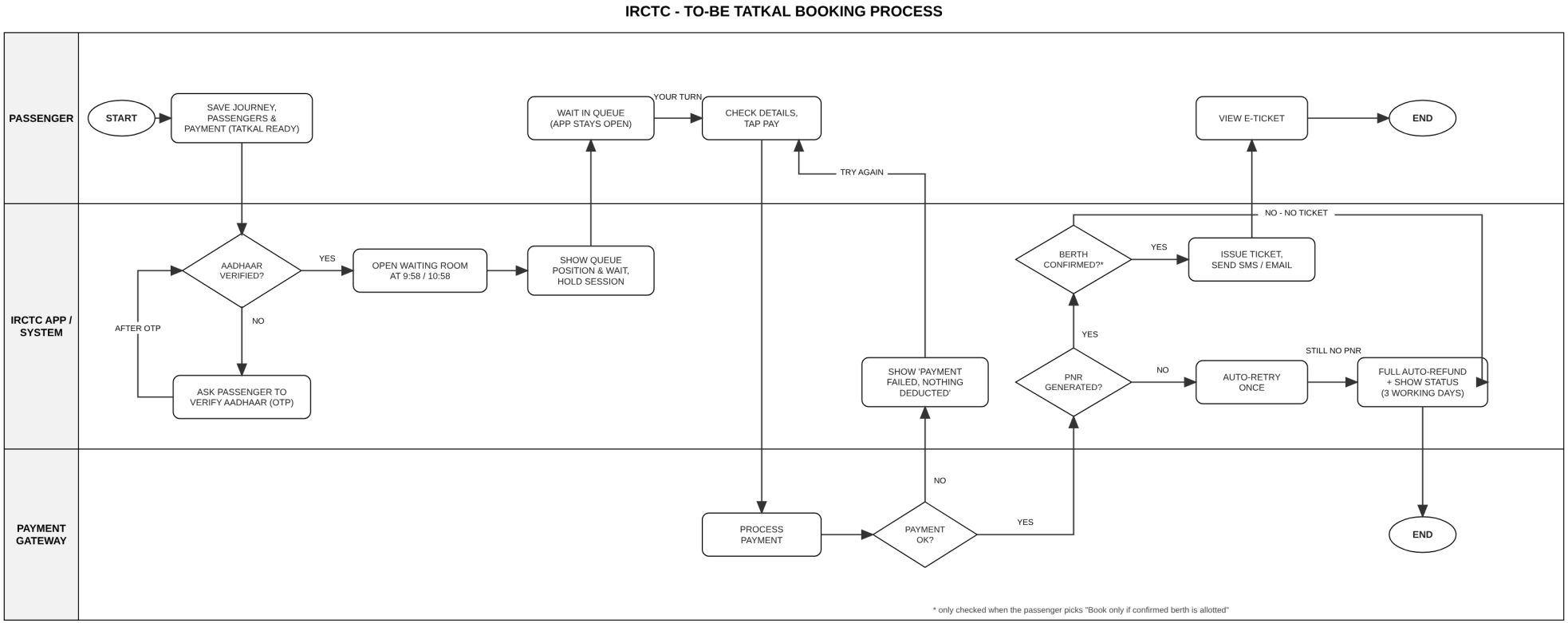
ID	Type	Requirement
NFR-01	Performance	Queue screen updates every 5 seconds; booking screens load within 2 seconds
NFR-02	Capacity	Waiting room handles the peak number of users at 10:00 without errors
NFR-03	Availability	Tatkal booking available 99.9% of the time between 9:55 and 11:30 AM
NFR-04	Security	Payment details stored as tokens only; Aadhaar data not stored in the app
NFR-05	Usability	Booking done in 3 taps after leaving the queue

Traceability (RTM)

BR	FR	Screen	Test case
BR-01	FR-01, FR-02, FR-03	WF-01	TC-01, TC-02, TC-03
BR-02	FR-04, FR-05	WF-02	TC-04, TC-05
BR-03	FR-06	WF-02, WF-03	TC-06
BR-04	FR-07	WF-03	TC-07
BR-05	FR-08, FR-09	WF-03, WF-04	TC-08, TC-09
BR-06	FR-10	WF-04	TC-10
BR-07	FR-11	WF-04	TC-11
BR-08	FR-12, FR-13	WF-05	TC-12
BR-09	FR-08, FR-14	WF-03	TC-08
BR-10	FR-15	-	-

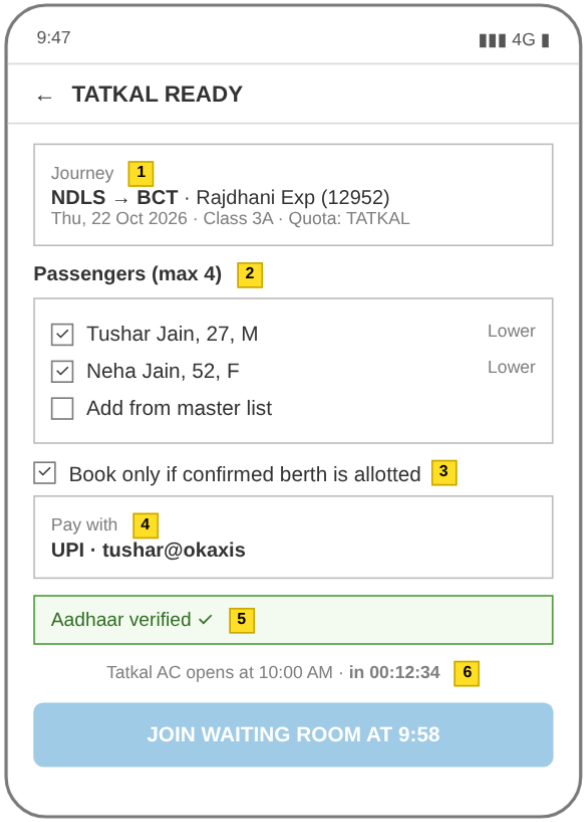
4. Process flow (To-Be)

Made in draw.io.



5. Wireframes

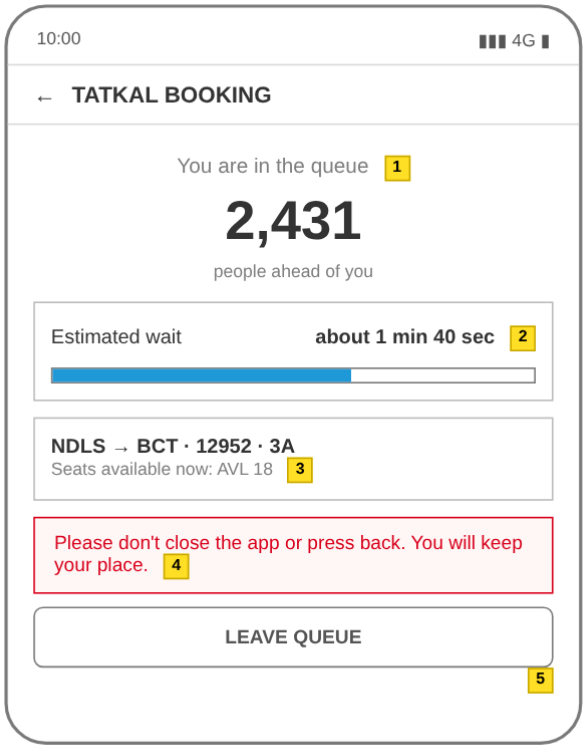
Made in Axure. Phone screens for the passenger, desktop for customer care. Yellow numbers match the notes.



WF-01 · Tatkal Ready · IRCTC Tatkal Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

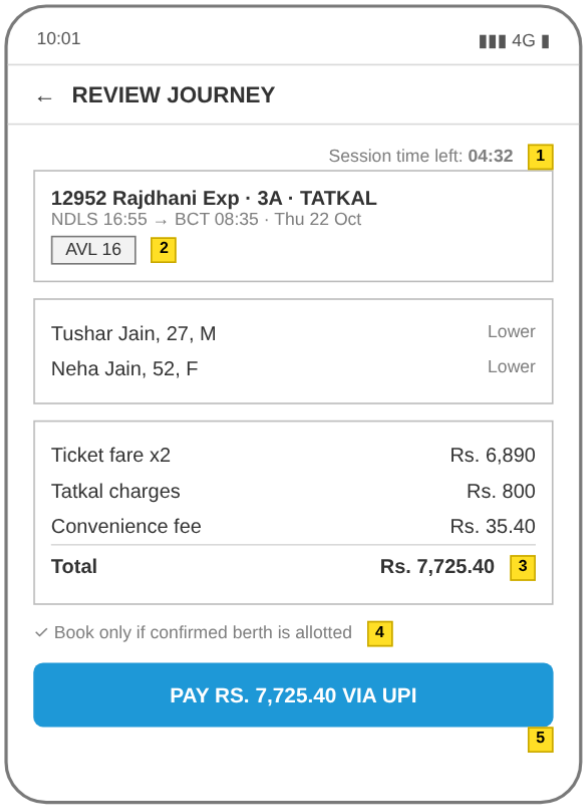
#	WIDGET	NOTE
1	**Journey card**	Train, date, class and quota picked the day before. Saved so the passenger doesn't type anything at 10:00 (FR-01).
2	**Passenger list**	Picked from the existing master list. Max 4 passengers per Tatkal PNR (RULE-01).
3	**Confirmed berth option**	If ticked, no ticket is booked unless a confirmed berth is allotted, and the full amount is refunded (RULE-06).
4	**Payment method**	Saved UPI ID or card, so payment is one tap later.
5	**Aadhaar status**	Tatkal needs an Aadhaar-verified account (RULE-02). If not verified, this shows 'Verify now' instead.
6	**Countdown**	Time left until the Tatkal window opens. Button turns active at 9:58 / 10:58 (RULE-03).



WF-02 · Waiting Room · IRCTC Tatkal Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

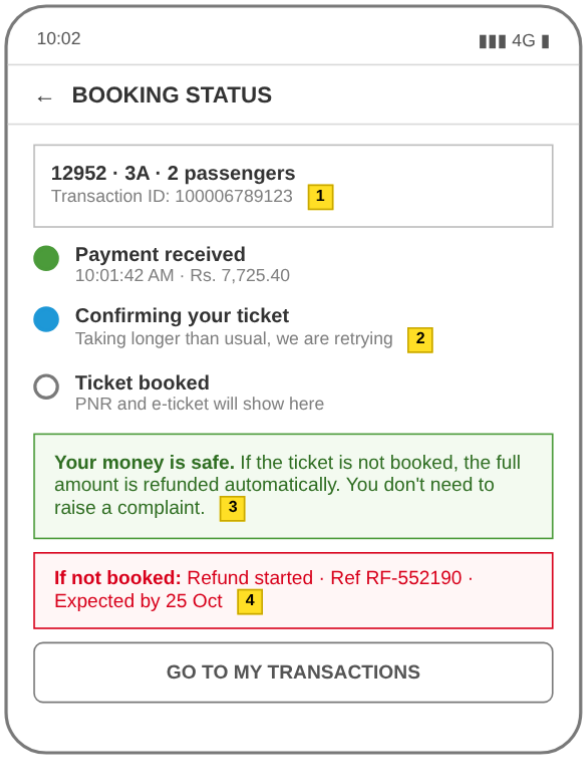
#	WIDGET	NOTE
1	**Queue position**	Replaces the old 'High load, please retry' error. Passenger sees their place in line and joins in the order they arrived (RULE-04).
2	**Estimated wait**	Calculated from how fast the queue is moving. Updates every 5 seconds (FR-04).
3	**Live availability**	Shows current seat status for the saved train and class so the passenger knows if it's worth waiting.
4	**Session warning**	Session is held while in the queue. The passenger is not logged out (RULE-05).
5	**Leave queue**	Asks 'Are you sure?' before leaving. Leaving gives up the place in the queue.



WF-03 · Review and Pay · IRCTC Tatkal Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

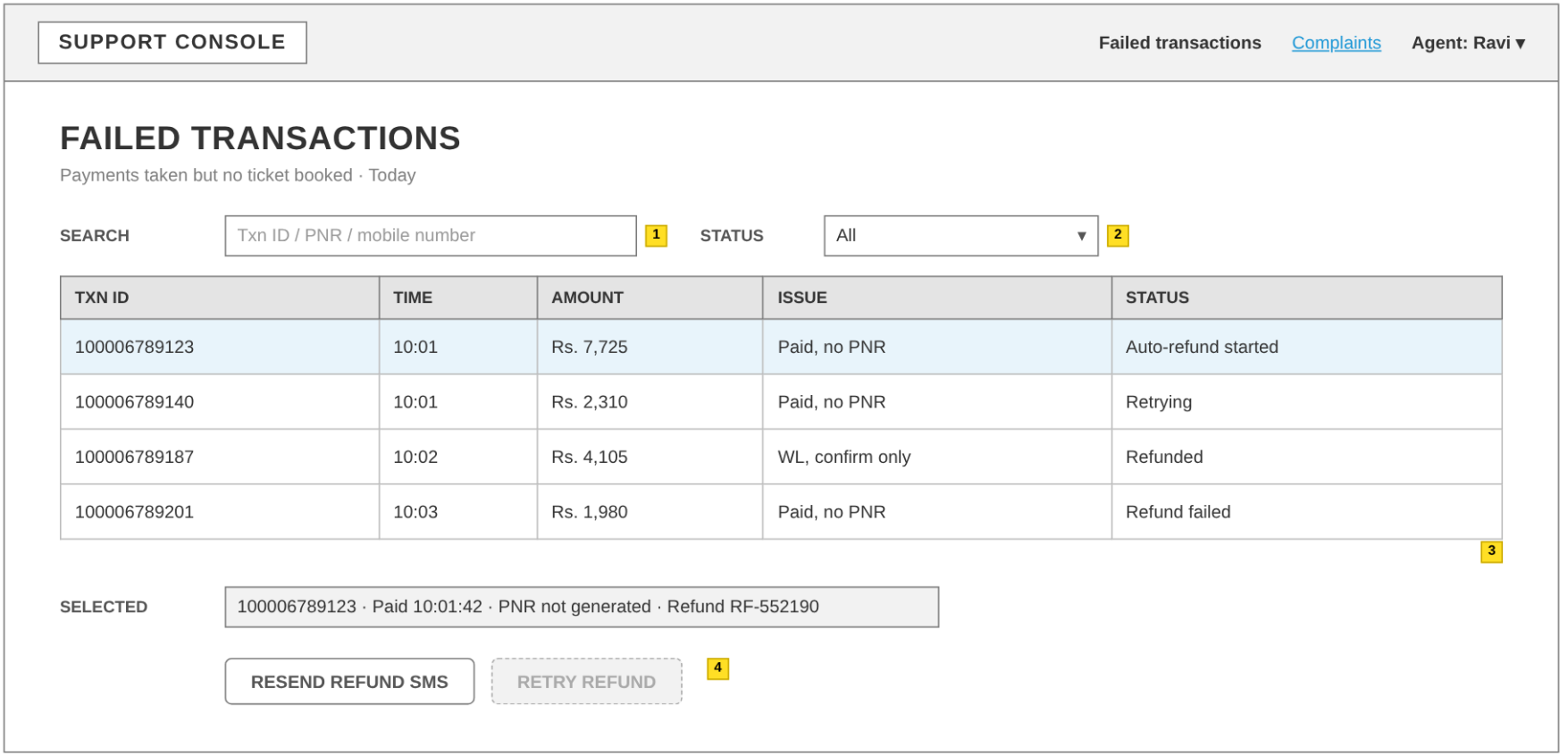
#	WIDGET	NOTE
1	**Session timer**	Passenger has 5 minutes to pay once their turn comes (RULE-05). At 1 minute left it turns red.
2	**Availability check**	Seat status checked again before payment. If it has changed to WL and the confirmed-berth option is on, the passenger is told before paying (FR-07).
3	**Fare breakup**	Full amount shown before payment, including Tatkal charges and convenience fee. Fares here are sample values.
4	**Confirmed berth option**	Carried over from Tatkal Ready. Can't be changed on this screen.
5	**Pay button**	Opens the saved UPI app. If payment fails, the passenger stays on this screen with 'Payment failed, nothing deducted' and can try again (FR-08).



WF-04 · Booking Status · IRCTC Tatkal Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

#	WIDGET	NOTE
1	**Transaction details**	Shown straight after payment so the passenger has a reference even if the app closes.
2	**Confirming step**	Replaces the old 'internal error' message. If no PNR within 2 minutes, the system retries once (RULE-07).
3	**Reassurance message**	Tells the passenger the money is safe and refund is automatic. Main fix for the 'money deducted, no ticket' complaints.
4	**Refund status**	Shown only if booking fails. Refund starts within 1 hour, credited within 3 working days (RULE-08). Push and SMS sent (FR-10).



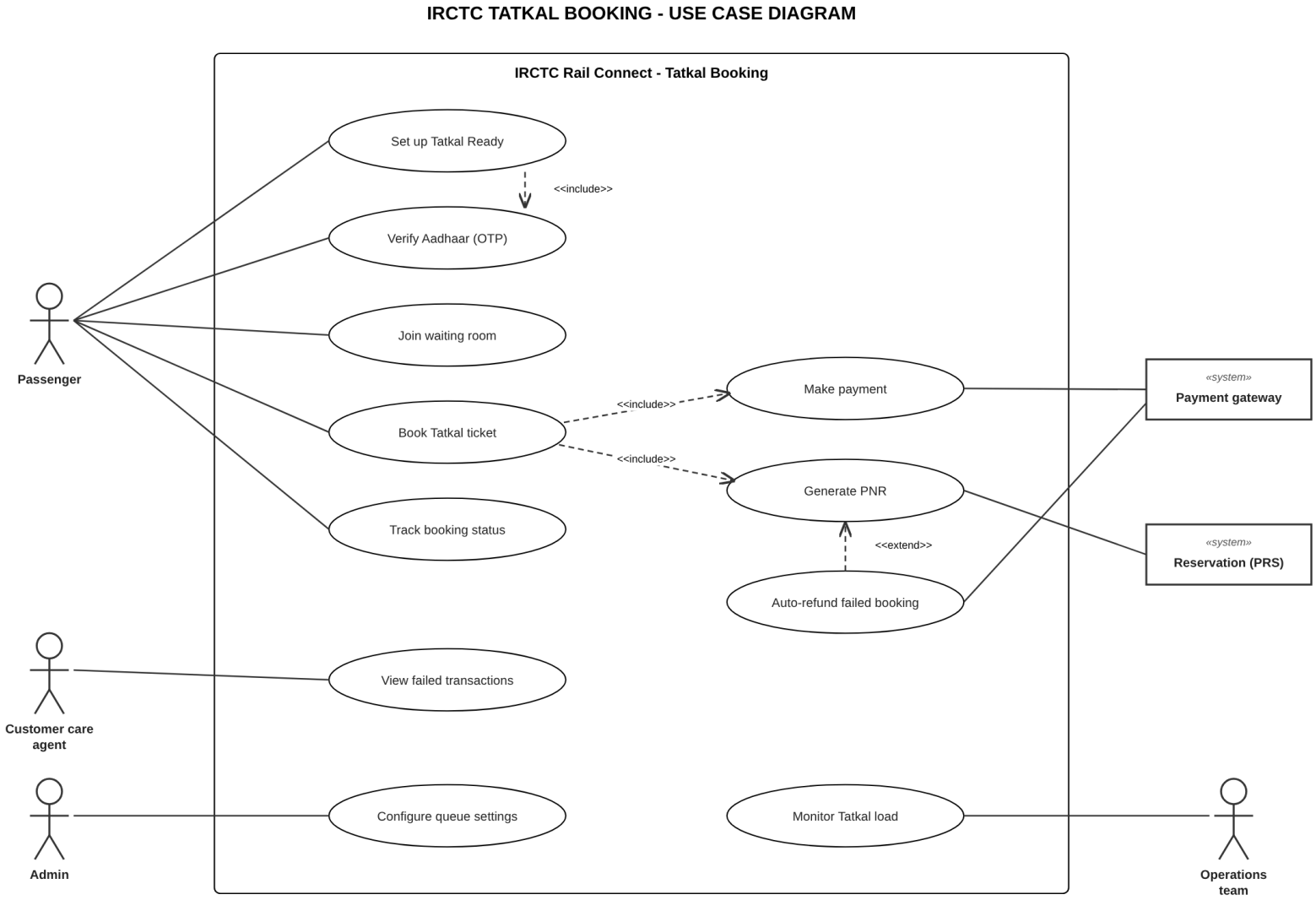
WF-05 · Failed Transactions (Customer Care) · IRCTC Tatkal Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

#	WIDGET	NOTE
1	**Search**	Agent can find a transaction by Txn ID, PNR or mobile number when a passenger calls.
2	**Status filter**	Retrying / Auto-refund started / Refunded / Refund failed.
3	**Transactions list**	All cases where payment was taken but no ticket was issued, newest first (FR-11). Most should already show 'Auto-refund started'.
4	**Retry refund / Resend SMS**	'Retry refund' is only active for 'Refund failed'. Every action is logged with the agent's name (FR-12).

6. Use case diagram

Made in draw.io.



7. User stories

For the Jira backlog. Points to be estimated by the dev team.

ID	Story	Acceptance criteria	FR
US-01	As a passenger, I want to save my Tatkal details the day before so I don't type anything at 10:00	Saved train, class, passengers and payment show on Tatkal Ready	FR-01, FR-02
US-02	As a passenger, I want to know if my Aadhaar isn't verified before Tatkal opens	Warning and 'Verify now' shown when saving	FR-03
US-03	As a passenger, I want to wait in a queue and see my place so I'm not stuck retrying	Queue number and wait time shown, updated every 5 sec	FR-04, FR-05
US-04	As a passenger, I don't want to be logged out while waiting	Session stays active in queue; 5 min to pay once in	FR-06
US-05	As a passenger, I want to know if seats changed before I pay	Warning shown if status changed to WL	FR-07
US-06	As a passenger, I want to know if my payment failed and nothing was taken	Clear message and Try again button	FR-08
US-07	As a passenger, I want to see what's happening after I pay	Status screen with each step and time	FR-09
US-08	As a passenger, I want my money back automatically if the ticket isn't booked	Refund ref and expected date shown and sent by SMS	FR-10
US-09	As a passenger, I want 'confirmed berth only' to really mean that	No WL ticket issued; full refund	FR-11
US-10	As a customer care agent, I want to see failed transactions so I can answer calls quickly	Search by Txn ID, PNR or mobile	FR-12, FR-13

8. UAT test cases

To be run by the app product team and customer care before launch.

ID	Test	Steps	Expected result	FR
TC-01	Save Tatkal Ready	Pick train, class, 2 passengers, UPI, save	Saved and shown on home screen	FR-01
TC-02	Add 5 passengers	Try to select a 5th passenger	Not allowed, 'max 4' message	FR-02
TC-03	Aadhaar not verified	Save Tatkal Ready with unverified account	'Verify now' shown, can't join queue	FR-03
TC-04	Join waiting room	Open app at 9:58	Queue number and wait time shown	FR-04
TC-05	Queue order	Two test users join 10 seconds apart	First joiner gets in first	FR-05
TC-06	Session in queue	Wait in queue for 3 minutes	Not logged out; 5-minute timer starts on entry	FR-06
TC-07	Seats change before payment	Seats go to WL while on review screen	Warning shown before pay	FR-07
TC-08	Payment fails	Cancel UPI payment	'Payment failed, nothing deducted' and Try again	FR-08
TC-09	Status after payment	Pay successfully	Status screen shows payment received and confirming	FR-09
TC-10	PNR not generated	Simulate no PNR after payment	Retry once, then refund started with ref and date	FR-10
TC-11	Confirmed berth only	Option on, only WL available	No ticket, full refund	FR-11
TC-12	Care sees failed txn	Search the Txn ID from TC-10	Shows 'Auto-refund started'	FR-12